

Telecom Customer Intelligence Dashboard

Revenue, Service Bundle & Churn Analytics

Cross-Sectional Subscriber Snapshot | Python + MySQL + Power BI Desktop

7,043 Customers · 26.54% Churn Rate · \$139,131/mo Revenue at Risk · 15× Contract Gap

TOOL	SOURCE	RECORDS	DATA TYPE
Python + MySQL + Power BI	IBM Sample Data (Kaggle)	7,043 · 29 Cols	Cross-Sectional (Single Snapshot)

01 EXECUTIVE SUMMARY

- Processed and cleaned 7,043 telecom subscriber records across geography, demographics, 9 service subscriptions, contract terms, and churn outcomes — achieving 100% data integrity by fixing an 11-row Total Charges text-formatting issue and dropping 4 constant/redundant columns from a 33-column raw source.
- Engineered 15 DAX measures (12 core KPI measures + 3 weighted-average correction measures) and built a 28-visual interactive Power BI dashboard featuring KPI cards, a geographic customer map, clustered bar/column charts, and 8 dynamic slicers enabling real-time cross-filtering across all dimensions.
- Analyzed a 26.54% overall churn rate (1,869 of 7,043 customers), surfacing contract type as the dominant lever — Month-to-month customers churn at 42.71% versus 2.83% for Two-year contracts, a 15× gap — and identified \$139,130.85 in monthly recurring revenue at risk (30.5% of total), annualizing to \$1,669,570.20.
- Uniquely leveraged IBM's extended dataset fields to validate a pre-built Churn Score (0.665 correlation with actual churn) and categorize 20 raw exit-interview reasons into 6 business categories, revealing Competitor Loss as the leading driver at 33.2% of all churn — ahead of Product/Price Dissatisfaction (25.6%) and Support/Attitude Problems (19.6%).

02 PROBLEM STATEMENT

A telecom provider maintained 7,043 subscriber records spanning demographics, 9 individual service subscriptions, contract terms, billing methods, and churn outcomes as a flat file with no structured reporting layer. Retention and revenue teams had no segmented view of which contract types, service bundles, or payment methods were driving the company's 26.54% churn rate, no quantified estimate of monthly recurring revenue exposed to churn, and — despite having both a pre-built predictive Churn Score and free-text exit reasons captured for every departing customer — no mechanism to validate the score's accuracy or translate raw exit reasons into an actionable, prioritized business narrative. Retention spend was being allocated without knowing whether competitive pressure, service gaps, or pricing was the primary driver behind departures.

03 OBJECTIVES

- Obj 1: Execute a structured Python EDA pipeline to clean the 33-column source, fix a text-formatted currency field, and engineer 4 boundary-safe derived features.

- Obj 2: Build a normalized MySQL database with inline data conversion, 8 performance indexes, and 8 validated analytical queries covering contract, tenure, service, geography, and churn-reason dimensions.
- Obj 3: Create a 13-dimension analytical VIEW enabling multi-factor Power BI segmentation across contract, tenure, CLTV, and charges simultaneously.
- Obj 4: Engineer DAX measures enabling dynamic churn rate calculation, revenue-at-risk quantification, and correctly weighted averages from the pre-aggregated VIEW.
- Obj 5: Build a 28-visual interactive dashboard covering churn overview, churn drivers, and customer segmentation across 3 pages, including a true geographic map using subscriber-level latitude/longitude.
- Obj 6: Surface actionable retention intelligence by categorizing 20 raw churn reasons into 6 business-relevant categories and validating IBM's pre-built Churn Score against actual outcomes, delivered with a portfolio-ready custom JSON theme.

04 DATASET OVERVIEW

Source & Structure

Attribute	Detail
Dataset Name	IBM Telco Customer Churn (Extended)
Source	Kaggle / IBM Sample Data Sets — Telco_customer_churn.xlsx
File Format	Excel Workbook (.xlsx) — richer than the standard 21-column CSV variant
Total Records	7,043 customers
Raw Columns	33 → 29 kept after dropping 4 constant/redundant fields
Data Coverage	Cross-sectional snapshot — no time dimension
Geographic Scope	1,129 unique cities, all within California (Country and State were constant and dropped)
Null Values	Zero, except Churn Reason (5,174 blank — correct, only churned customers have an exit reason)
Data Quality Issue	Total Charges stored as text with 11 blank values, all belonging to customers with Tenure = 0 (brand-new sign-ups)

Column Definitions (Key Fields)

Column	Type	Description
customer_id	Text (PK)	Unique subscriber identifier
city, zip_code, latitude, longitude	Geo	Enables true geographic mapping — unique to this dataset vs the standard CSV variant
senior_citizen, partner, dependents	Text → Converted	senior_citizen converted Yes/No → 1/0 at import
tenure_months	Whole Number	Months as a subscriber (0–72)
9 service fields	Text	Phone, Internet, Online Security/Backup, Device Protection, Tech Support, Streaming TV/Movies, Multiple Lines

Column	Type	Description
contract, payment_method	Text	Month-to-month/One year/Two year; 4 payment methods
monthly_charges, total_charges	Decimal	Billing amounts in USD
churn_value, churn_score, cltv, churn_reason	Mixed	Actual churn flag, IBM's predictive score (0–100), lifetime value, and free-text exit reason — all unique to the extended dataset

05 TOOLS & TECHNOLOGIES USED

Tool / Technology	Platform	Purpose in This Project
Python 3.11	pandas, numpy, matplotlib, seaborn, openpyxl	EDA, XLSX ingestion, feature engineering, 10-chart visualization, CSV export
MySQL 8.0	MySQL Workbench	Table creation, LOAD DATA INFILE import, indexing, 8 analytical queries, VIEW creation
Power BI Desktop	Microsoft (Latest 2024)	Primary BI tool for MySQL connection, DAX modelling, and 28-visual dashboard including geographic map
DAX (Data Analysis Expressions)	Power BI Native Language	15 measures: 12 core KPIs + 3 weighted-average correction measures
IBM Sample Data Sets	Public Dataset Platform	Free extended Telco churn dataset with geography, CLTV, and churn reasons
JSON Theme File	Power BI Custom Theme	Telecom Signal Intelligence — Network Blue portfolio theme

06 DATA PREPARATION — PYTHON & MYSQL PIPELINE

Raw XLSX contained 4 zero-value or redundant columns, a text-formatted currency field with 11 blank values, and — discovered during MySQL analysis — a hidden carriage-return artifact affecting one query's text categorization. All steps below were executed in sequence, creating a repeatable, verifiable pipeline from raw XLSX through to a query-ready MySQL database.

Phase 1 — Python (10 Steps)

#	Python Action	Reason / Impact
01	pd.read_excel('Telco_customer_churn.xlsx')	Confirmed shape (7043, 33), zero unexpected nulls
02	Compared XLSX (33 cols) against the standard companion CSV (21 cols)	Selected XLSX as primary source — strictly richer with geography, CLTV, Churn Score, Churn Reason
03	Dropped Count, Country, State, Lat Long columns	All constant or redundant — reduced to 29 meaningful columns
04	Converted Total Charges from text to numeric, filled 11 blanks with 0	Confirmed all 11 blanks belonged to Tenure=0 customers — correct treatment

#	Python Action	Reason / Impact
05	Engineered TenureBucket, CLTVBucket, ChargesBucket, ChurnReasonCategory	4 derived features; reason category groups 20 raw text reasons into 6 business buckets
06	Generated Charts 1–3 (Overview, Contract, Tenure)	Confirmed 26.54% churn and the 15× contract-type gap
07	Generated Charts 4–6 (Internet, Payment, Reason Category)	Surfaced Fiber Optic and Electronic Check as high-risk segments; validated reason categorization
08	Generated Charts 7–9 (Scatter, Service Add-Ons, Score Validation)	Confirmed IBM's Churn Score correlates at 0.665 with actual churn
09	Generated Chart 10 (Correlation Heatmap) and fixed a pie-chart unpacking error	Confirmed Churn Score (0.665) and Tenure (-0.352) as dominant correlates
10	df_export.to_csv('telecom_cleaned.csv')	Exported 7,043 clean rows, 29 columns, ready for MySQL

Phase 2 — MySQL (13 Steps)

#	MySQL Action	Reason / Impact
01	CREATE TABLE telecom_churn (29 columns, customer_id PK)	Normalized schema across geography, services, billing, and churn fields
02	LOAD DATA INFILE with SET senior_citizen = IF(@raw='Yes',1,0)	7,043 rows loaded, 0 skipped, 0 warnings
03	Verified row count and senior_citizen/churn_value distributions	Confirmed 7,043 total, 1,869 churned matching Python exactly
04	Created 8 indexes (churn, contract, internet, payment, tenure, senior, city, tech support)	Optimized analytical query performance
05	Query 1 — Overall KPIs	26.54% churn, \$139,130.85 monthly revenue at risk, avg CLTV \$4,400
06	Query 2 — Churn by Contract Type	Month-to-month 42.71% → Two-year 2.83% (15× gap)
07	Query 3 — Internet Service × Payment Method combination	Fiber Optic + Electronic Check is the single worst combination at 53.23% churn
08	Query 4 — Churn by Tenure Bucket	0-6 months: 52.94% churn → 49-72 months: 9.51%
09	Query 5 — Service Add-Ons Impact	Tech Support cuts churn from 41.64% to 15.17%
10	Query 6 — Top Cities by Churn Rate (20+ customers)	Santa Rosa highest at 45.83%; Los Angeles (largest city) at 29.51%
11	Query 7 — Churn Reason Category Breakdown	Initially returned 100% in one category due to a hidden carriage-return character; diagnosed and fixed via TRIM(TRAILING 'r' FROM churn_reason) (see Challenge 3)
12	CREATE VIEW vw_telecom_summary	13-dimension aggregation for Power BI segmentation
13	Connected Power BI to MySQL, renamed both tables in Power Query	Removed database-prefix naming so DAX measures resolve correctly

07 DAX MEASURES CREATED

15 DAX measures were created via right-click on table → New Measure. As with the prior MySQL-sourced project in this portfolio, 1 weighted-average measure was proactively built alongside the core 12 — anticipating that vw_telecom_summary's pre-aggregated columns cannot be safely SUM'd across a 13-dimension VIEW.

Measure Name	DAX Formula	Result / Use
Total Customers	COUNTROWS(telecom_churn)	7,043 — primary KPI card
Total Churned	SUM(telecom_churn[churn_value])	1,869 — headline KPI
Churn Rate %	DIVIDE([Total Churned],[Total Customers])*100	26.54%
Revenue At Risk	CALCULATE([Total Monthly Revenue], churn_value=1)	\$139,130.85/month
Revenue At Risk %	DIVIDE([Revenue At Risk],[Total Monthly Revenue])*100	30.50%
Month to Month Churn Rate	CALCULATE([Churn Rate %], contract="Month-to-month")	42.71% — headline driver KPI
Avg Churn Score	AVERAGE(telecom_churn[churn_score])	58.70 — validates against 0.665 correlation
VIEW Weighted Churn Rate	DIVIDE(SUMX(view,rate*count),SUM(count))	Corrects VIEW aggregation across all tenure/CLTV/charges-segmented visuals

08 DASHBOARD VISUALS BUILT (28 Visuals)

Page 1 — Churn Overview (11 Visuals)

#	Visual Type	Fields Used	Insight Delivered
1	KPI Card	Total Customers measure	7,043 total subscriber base
2	KPI Card	Total Churned measure	1,869 customers churned
3	KPI Card	Churn Rate % measure	26.54% overall churn rate
4	KPI Card	Revenue At Risk measure	\$139,131 monthly revenue at risk
5	KPI Card	Avg CLTV measure	\$4,400 average customer lifetime value
6	KPI Card	Avg Churn Score measure	58.70 average predictive churn score
7	Donut Chart	churn_label / Total Customers measure	73.46% Retained vs 26.54% Churned
8	Bar Chart	contract / Churn Rate % measure	Month-to-month leads at 42.71%
9	Map Visual	latitude, longitude / Total Customers, Churn Rate %	Geographic churn distribution across 1,129 California cities
10	Slicer	internet_service field	Filters all visuals by internet service type
11	Slicer	payment_method field	Filters all visuals by payment method

Page 2 — Churn Drivers (8 Visuals)

#	Visual Type	Fields Used	Insight Delivered
12	Bar Chart	tenure_bucket / VIEW Weighted Churn Rate	0-6 months highest at 52.94% churn
13	Column Chart	internet_service / Churn Rate % measure	Fiber Optic churns 2x more than DSL
14	Bar Chart	payment_method / Churn Rate % measure	Electronic Check is highest-risk payment method
15	Clustered Column	tech_support / Churn Rate % measure	Tech Support nearly halves churn risk
16	Donut Chart	Churn Reason Category (6 buckets)	Competitor Loss leads at 33.2% of all churn
17	Column Chart	churn_label / Avg Churn Score measure	Score validation: 82.5 (churned) vs 50.1 (retained)
18	Slicer	contract field	Filters by contract type
19	Slicer	tenure_bucket field (VIEW)	Filters by tenure bracket

Page 3 — Customer Segmentation (9 Visuals)

#	Visual Type	Fields Used	Insight Delivered
20	KPI Card	Month to Month Churn Rate measure	42.71% confirmed as headline driver
21	KPI Card	Revenue At Risk % measure	30.50% of monthly revenue at risk
22	KPI Card	Avg Monthly Charges measure	\$64.76 average monthly bill
23	KPI Card	Avg Tenure measure	32.4 months average subscriber tenure
24	Column Chart	senior_status (VIEW) / VIEW Weighted Churn Rate	Senior citizens churn nearly 2x more
25	Column Chart	dependents / Churn Rate % measure	Having dependents = 5x lower churn risk
26	Table	city, total_customers, churn_rate_pct (20+ customer cities)	Santa Rosa highest-risk city at 45.83%
27	Slicer	senior_citizen field	Filters by senior citizen status
28	Slicer	dependents field	Filters by dependents status

09 KEY INSIGHTS & RESULTS

A. Contract Type — The Dominant Retention Lever

- Month-to-month customers churn at 42.71% versus 11.27% for One-year and just 2.83% for Two-year contracts — a 15× gap between the highest and lowest-risk contract tiers, on a base of 3,875 / 1,473 / 1,695 customers respectively.
- The combination of Fiber Optic internet paid via Electronic Check is the single worst-performing segment in the entire dataset at 53.23% churn (849 of 1,595 customers) — nearly double the overall average and confirming that risk factors compound rather than simply add.

B. Tenure — The First 6 Months Are Critical

- Customers in their first 6 months churn at 52.94% — over half leave before establishing any meaningful relationship with the service — falling to 35.89% at 7-12 months, 28.71% at 13-24 months, and just 9.51% by 49-72 months.
- Average monthly charges rise steadily with tenure (\$54.74 in months 0-6 to \$73.95 in months 49-72), meaning the least loyal segment is also being charged the least — an onboarding and early-engagement gap rather than a pricing problem.

C. Service Bundle Impact

- Every protective service add-on nearly halves churn risk: Online Security (41.77% → 14.61%), Tech Support (41.64% → 15.17%), Online Backup (39.93% → 21.53%), and Device Protection (39.13% → 22.50%) all show the same pattern.
- Customers with no internet service at all churn at just 7.40% — the lowest rate of any segment — confirming that churn risk is concentrated specifically among internet subscribers without protective add-ons, not internet service itself.

D. Why Customers Actually Leave

- Categorizing 20 raw exit reasons into 6 business buckets reveals Competitor Loss as the leading driver at 33.2% of all churn (621 of 1,869), followed by Product/Price Dissatisfaction at 25.6% (478) and Support/Attitude Problems at 19.6% (366).
- Network Issues (5.5%) and Self-Service Gaps (4.7%) are comparatively minor — indicating retention investment is better directed at competitive positioning and support quality than infrastructure reliability.

E. Predictive Score Validation & Demographics

- IBM's pre-built Churn Score validates strongly against actual outcomes — churned customers average 82.51 versus 50.10 for retained customers, a 0.665 correlation coefficient that exceeds every other individual variable in the dataset.
- Having dependents is the single strongest demographic protective factor (6.52% churn vs 32.55% without) — a 5× gap — while senior citizens churn nearly twice as often as non-seniors (41.68% vs 23.61%).

KPI Summary

KPI Metric	Value	KPI Metric	Value
Total Customers	7,043	Total Churned	1,869 (26.54%)
Month-to-Month Churn	42.71%	Two-Year Churn	2.83% (15× gap)

KPI Metric	Value	KPI Metric	Value
0-6 Month Tenure Churn	52.94%	49-72 Month Tenure Churn	9.51%
Fiber Optic Churn	41.89%	DSL Churn	18.96%
Electronic Check Churn	45.29%	Credit Card (Auto) Churn	15.24%
Worst Combination	Fiber Optic + Electronic Check — 53.23% churn (849 of 1,595)		
Monthly Revenue at Risk	\$139,130.85	Annualized Revenue Loss	\$1,669,570.20
Top Churn Reason	Competitor Loss — 33.2%	2nd Reason	Product/Price — 25.6%
Churn Score — Churned	82.51	Churn Score — Retained	50.10
Has Dependents Churn	6.52%	No Dependents Churn	32.55% (5× gap)
Correlation — Churn Score	0.665	Correlation — Tenure	-0.352

10 CHALLENGES FACED & HOW OVERCOME

Challenge 1 — Missing openpyxl Dependency

- Problem: `pd.read_excel()` raised `ImportError: Missing optional dependency 'openpyxl'` — pandas requires this separate library to read .xlsx files, and it wasn't part of the environment from prior CSV-only projects.
- Solution: Installed via `python -m pip install openpyxl`, resolving the dependency without any code changes required.

Challenge 2 — Matplotlib Pie Chart Return-Value Mismatch

- Problem: `ValueError: not enough values to unpack (expected 3, got 2)` on the Churn Reason donut chart. `ax.pie()` only returns a third value (autotexts) when the `autopct` parameter is supplied — this chart used custom pre-formatted labels instead, so only 2 values were actually returned.
- Solution: Changed `wedges, texts, autotexts = ax.pie(...)` to `wedges, texts = ax.pie(...)` — verified via isolated reproduction before applying to the full script.

Challenge 3 — Hidden Carriage Return Broke Text Categorization

- Problem: A SQL query categorizing churn reasons into 6 groups returned 100% of rows in a single catch-all category. Root cause: the CSV was exported with Windows-style `\r\n` line endings, but `LOAD DATA INFILE` was configured with `LINES TERMINATED BY '\n'` only — leaving a trailing `\r` attached to `churn_reason`, the last column in the row. MySQL's default `TRIM()` only strips spaces, not carriage returns, so the hidden character survived import undetected.
- Solution: Diagnosed by comparing `LENGTH()` of stored values against expected string lengths, then corrected with `UPDATE telecom_churn SET churn_reason = TRIM(TRAILING '\r' FROM churn_reason)` — a single statement that fixed all 1,869 affected rows without requiring re-import. The bug was isolated to one query; the VIEW and Power BI dashboard never referenced this field and were unaffected.

Challenge 4 — VIEW Aggregation Risk (Proactively Addressed)

- Problem: Based on an identical issue encountered in a prior MySQL-sourced project in this portfolio, any Power BI visual built directly on vw_telecom_summary's pre-aggregated rate/average columns using default SUM aggregation would display inflated, meaningless totals.
- Solution: Built VIEW Weighted Churn Rate proactively alongside the 12 core measures — before any visual was built rather than after discovering the bug — using $\text{SUMX}(\text{view}[\text{rate}] \times \text{view}[\text{count}]) / \text{SUM}(\text{view}[\text{count}])$ to guarantee correctly weighted results from the first dashboard draft.

11 SKILLS LEARNED & TAKEAWAYS

- Multi-Format Source Evaluation: Compared two versions of the same public dataset (21-column CSV vs 33-column XLSX) before starting analysis, and selected the strictly richer source — a habit that materially expanded the project's analytical scope to include geography and validated predictive scoring.
- Invisible-Character Debugging: Diagnosed a silent data corruption bug (trailing carriage return) that produced no error message and no warning during import, only surfacing as a suspiciously uniform query result — reinforcing that a 100%-in-one-bucket output is itself a diagnostic signal worth investigating, not a data pattern to accept at face value.
- Cross-Tool Verification Discipline: Independently verified that Python's churn reason categorization (621/478/366/213/103/88) and MySQL's post-fix Query 7 produced identical results — confirming the fix was correct rather than merely no-longer-erroring.
- Validating External Predictive Scores: Treated IBM's pre-built Churn Score as a hypothesis to test rather than a given — quantifying its 0.665 correlation with actual outcomes before featuring it as a dashboard KPI, rather than assuming a vendor-supplied score is automatically trustworthy.
- Proactive Bug Prevention: Applied a lesson from a prior project's VIEW-aggregation bug before it recurred here — building the weighted-average measure into the initial DAX layer rather than discovering and patching the same class of bug a second time.

12 CONCLUSION

- Processed 7,043 telecom subscriber records through a structured Python + MySQL pipeline — selecting the richer of two available source formats, fixing a text-formatted currency field, engineering 4 derived features, and building a 29-column normalized database with 8 validated analytical queries.
- Engineered 15 DAX measures and built a 28-visual interactive dashboard spanning KPI cards, a true geographic map, comparison bar/column charts, a churn-reason donut, and 8 cross-filtering slicers across 3 dashboard pages.
- Surfaced contract type as the dominant retention lever (15× churn gap), quantified \$139,130.85 in monthly revenue at risk, validated IBM's predictive Churn Score at 0.665 correlation, and categorized raw exit reasons to reveal Competitor Loss (33.2%) as the leading actual driver of churn — insights unavailable in the standard 21-column version of this dataset.
- Delivered a portfolio-grade custom JSON theme (Telecom Signal Intelligence — Network Blue) demonstrating end-to-end command of Python, MySQL, and Power BI from raw multi-format ingestion through ETL, DAX modelling, invisible-character debugging, and professional dashboard theming.